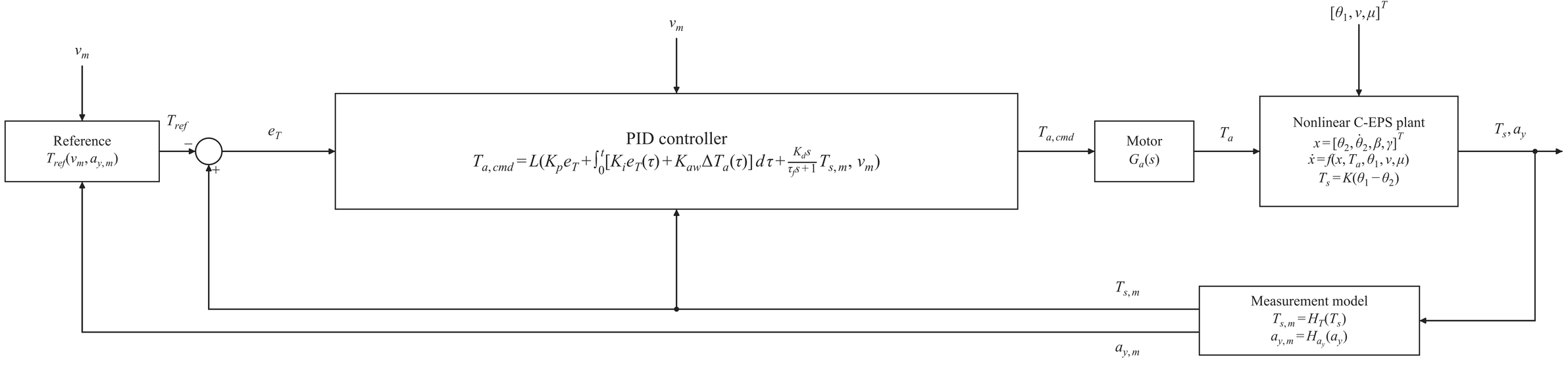
$v_m$
Reference
$T_{ref}(v_m, a_{y,m})$
$T_{ref}$
−
+
$e_T$
$v_m$
PID controller
$T_{a,cmd} = L(K_p e_T + \int_0^t [K_i e_T(\tau) + K_{aw} \Delta T_a(\tau)]\, d\tau + \frac{K_d s}{\tau_f s + 1} T_{s,m},\ v_m)$
$T_{a,cmd}$
Motor
$G_a(s)$
$T_a$
$[\theta_1, v, \mu]^T$
Nonlinear C-EPS plant
$x = [\theta_2, \dot{\theta}_2, \beta, \gamma]^T$
$\dot{x} = f(x, T_a, \theta_1, v, \mu)$
$T_s = K(\theta_1 - \theta_2)$
$T_s, a_y$
$T_{s,m}$
$a_{y,m}$
Measurement model
$T_{s,m} = H_T(T_s)$
$a_{y,m} = H_{a_y}(a_y)$